

For the use only of a Registered Medical Practitioner or a Hospital or a Laboratory

**Bortezomib Kabi
Powder for Solution for Injection
3.5 mg/vial**

1. NAME OF THE MEDICINAL PRODUCT

Bortezomib Kabi powder for solution for injection 3.5 mg/vial

2. QUALITATIVE AND QUANTITATIVE COMPOSITION

Each vial contains 3.5 mg bortezomib (as a mannitol boronic ester).
After reconstitution, 1 ml of solution for subcutaneous injection contains 2.5 mg bortezomib. After reconstitution, 1 ml of solution for intravenous injection contains 1 mg bortezomib.

For the full list of excipients, see section 6.1.

3. PHARMACEUTICAL FORM

Powder for solution for injection.
White to off-white lyophilized powder or cake.

4. CLINICAL PARTICULARS

4.1 Therapeutic indications

Bortezomib is indicated for the treatment of patients with multiple myeloma.

Bortezomib is indicated for the treatment of patients with mantle cell lymphoma.

4.2 Posology and method of administration

Important Dosing Guidelines

Bortezomib is for intravenous or subcutaneous use only. Do not administer bortezomib with any other route.

Because each route of administration has a different reconstituted concentration, use caution when calculating the volume to be administered.

The recommended starting dose of bortezomib is 1.3 mg/m². Bortezomib may be administered intravenously at a concentration of 1 mg/ml, or subcutaneously at a concentration of 2.5 mg/ml. Bortezomib retreatment may be considered for patients with multiple myeloma who had previously responded to treatment with bortezomib and who have relapsed at least 6 months after completing prior bortezomib treatment. Treatment may be started at the last tolerated dose.
When administered intravenously, administer bortezomib as a 3 to 5 second bolus intravenous injection.

Dosage in Previously Untreated Multiple Myeloma

Bortezomib is administered in combination with oral melphalan and oral prednisone for 9, six-week treatment cycles as shown in **Table 1**. In Cycles 1 to 4, bortezomib is administered twice weekly (Days 1, 4, 8, 11, 22, 25, 29 and 32). In Cycles 5 to 9, bortezomib is administered once weekly (Days 1, 8, 22 and 29). At least 72 hours should elapse between consecutive doses of Bortezomib.

Table 1: Dosage Regimen for Patients with Previously Untreated Multiple Myeloma

Twice Weekly Bortezomib (Cycles 1 to 4)													
Week	1	2	3	4	5	6							
Bortezomib (1.3 mg/m ²)	Day 1	--	--	Day 8	Day 11	rest period	Day 22	Day 25	Day 29	Day 32	rest period		
Melphalan (0.9 mg/m ²)	Day 1	Day 2	3	4	--	--	rest period	--	--	--	--		
Prednisone (60 mg/m ²)	1	2	3	4	--	--	rest period	--	--	--	--		

Once Weekly Bortezomib (Cycles 5 to 9 when used in combination with Melphalan and Prednisone)													
Week	1	2	3	4	5	6							
Bortezomib (1.3 mg/m ²)	Day 1	--	--	Day 8	Day 22	Day 29	rest period	Day 1	--	--	Day 8	Day 22	Day 29
Melphalan (0.9 mg/m ²)	Day 1	Day 2	3	4	--	--	rest period	Day 1	Day 2	3	4	--	--
Prednisone (60 mg/m ²)	1	2	3	4	--	--	rest period	1	2	3	4	--	--

Dose Modification Guidelines for Bortezomib when given in Combination with Melphalan and Prednisone
Prior to initiating any cycle of therapy with Bortezomib in combination with melphalan and prednisone:

- Platelet count should be at least 70 x 10⁹/L and the absolute neutrophil count (ANC) should be at least 1.0 x 10⁹/L.
- Nonhematological toxicities should have resolved to Grade 1 or baseline.

Table 2: Dose Modifications during Cycles of Combination Bortezomib, Melphalan and Prednisone Therapy

Toxicity	Dose Modification or Delay
Hematological toxicity during a cycle: If prolonged Grade 4 neutropenia or thrombocytopenia, or thrombocytopenia with bleeding is observed in the previous cycle If platelet count is not above 30 x 10 ⁹ /L or ANC is not above 0.75 x 10 ⁹ /L on a Bortezomib dosing day (other than day 1) If several bortezomib doses in consecutive cycles are withheld due to toxicity Grade 3 or higher non-hematological toxicities	Consider reduction of the melphalan dose by 25% in the next cycle Withhold bortezomib dose Reduce bortezomib dose by 1 dose level (from 1.3 mg/m ² to 1 mg/m ² or from 1 mg/m ² to 0.7 mg/m ²) Withhold bortezomib therapy until symptoms of toxicity have resolved to Grade 1 or baseline. Then, bortezomib may be reinitiated with one dose level reduction (from 1.3 mg/m ² to 1 mg/m ² , or from 1 mg/m ² to 0.7 mg/m ²). For bortezomib - related neuropathic pain and/or peripheral neuropathy, hold or modify bortezomib as outlined in Table 6.

For dose modifications guidelines for peripheral neuropathy, see Dose modifications for peripheral neuropathy below.

Dose Modifications for Peripheral Neuropathy

Starting bortezomib subcutaneously may be considered for patients with pre-existing or at high risk of peripheral neuropathy. When rituximab with pre-existing severe neuropathy should be treated with bortezomib only after careful risk-benefit assessment.

Patients experiencing new or worsening peripheral neuropathy during bortezomib therapy may require a decrease in the dose and/or a less dose-intense schedule.

For dose or schedule modification guidelines for patients who experience bortezomib-related neuropathic pain and/or peripheral neuropathy see **Table 6**.

Table 3: Posology for Bortezomib combination therapy for patients with previously untreated multiple myeloma eligible for haematopoietic stem cell transplantation

Combination therapy with dexamethasone													
Bortezomib 3.5mg powder for solution for injection is administered via intravenous or subcutaneous injection at the recommended dose of 1.3 mg/m ² body surface area twice weekly for two weeks on days 1, 4, 8, and 11 in a 21-day treatment cycle. This 3-week period is considered a treatment cycle. At least 72 hours should elapse between consecutive doses of bortezomib. Dexamethasone is administered orally at 40 mg on days 1, 2, 3, 4, 8, 9, 10 and 11 of the bortezomib treatment cycle.													
Four treatment cycles of this combination therapy are administered.													

Combination therapy with dexamethasone and thalidomide													
Bortezomib 3.5mg powder for solution for injection is administered via intravenous or subcutaneous injection at the recommended dose of 1.3 mg/m ² body surface area twice weekly for two weeks on days 1, 4, 8, and 11 in a 21-day treatment cycle. This 3-week period is considered a treatment cycle. At least 72 hours should elapse between consecutive doses of bortezomib. Dexamethasone is administered orally at 40 mg on days 1, 2, 3, 4, 8, 9, 10 and 11 of the bortezomib treatment cycle.													
Thalidomide is administered orally at 50 mg daily on days 1-14 and if tolerated the dose is increased to 100 mg on days 15-28, and thereafter may be further increased to 200 mg daily from cycle 2 (see Table 3).													
Four treatment cycles of this combination are administered. It is recommended that patients with at least partial response receive 2 additional cycles.													

Table 3: Posology for Bortezomib combination therapy for patients with previously untreated multiple myeloma eligible for haematopoietic stem cell transplantation

B-Dx					
Cycles 1 to 4					
Week	1	2	3		
B (1.3 mg/m ²)	Day 1, 4	Day 8, 11	Rest Period		
Dx 40 mg	Day 1, 2, 3, 4	Day 8, 9, 10, 11	-		
B-Dx+T					
Cycles 1 to 4					
Week	1	2	3	4	
B (1.3 mg/m ²)	Day 1, 4	Day 8, 11	Rest Period	Rest Period	
T 50 mg	Daily	Daily	-	-	
T 100 mg*	-	-	Daily	Daily	
Dx 40 mg	Day 1, 2, 3, 4	Day 8, 9, 10, 11	-	-	
Cycles 2 to 4*					
B (1.3 mg/m ²)	Day 1, 4	Day 8, 11	Rest Period	Rest Period	
T 200 mg*	Daily	Daily	Daily	Daily	
Dx 40 mg	Day 1, 2, 3, 4	Day 8, 9, 10, 11	-	-	

Cycles 2 to 4*
B (1.3 mg/m²)
T 50 mg
Dx 40 mg
Day 1, 4
Day 8, 11
Rest Period
Rest Period

B+Bortezomib; Dx=dexamethasone; T=thalidomide

*Thalidomide dose is increased to 100 mg from week 3 of Cycle 1 only if 50 mg is tolerated and to 200 mg from cycle 2 onwards if 100 mg is tolerated.

*Up to 6 cycles may be given to patients who achieve at least a partial response after 4 cycles

Dosage adjustments for transplant eligible patients

For bortezomib dosage adjustments, as described under **"Dosage and Dose Modifications for Relapsed Multiple Myeloma and Related Mantle Cell Lymphoma"** and **"Dose Modifications for Peripheral Neuropathy"** should be followed.
In addition, when bortezomib is given in combination with other chemotherapeutic medicinal products, appropriate dose reductions for these products should be considered in the event of toxicities according to the recommendations in the local package inserts.

Dosage in Previously Untreated Mantle Cell Lymphoma

Bortezomib (1.3 mg/m²) is administered intravenously in combination with intravenous rituximab, cyclophosphamide, doxorubicin and oral prednisone (VCR-CAP) for 6 three week treatment cycles as shown in **Table 4**. Bortezomib is administered first followed by rituximab. Bortezomib is administered twice weekly for two weeks (Days 1, 4, 8, and 11) followed by a ten day rest period. On Days 12 to 21. For patients with a response first documented at cycle 6, two additional VCR-CAP cycles are recommended. At least 72 hours should elapse between consecutive doses of bortezomib.

Table 4: Dosage Regimen for Patients with Previously Untreated Mantle Cell Lymphoma													
Twice Weekly Bortezomib (6, Three Week Cycles)*													
Week	1	2	3										
Bortezomib (1.3 mg/m ²)	Day 1	--	--	Day 4	--	--	Day 8	Day 11	Rest period				
Rituximab (375 mg/m ²)	Day 1	--	--	rest period	--	--	--	--	Rest period				
Cyclophosphamide (750 mg/m ²)	Day 1	--	--	rest period	--	--	--	--	Rest period				
Doxorubicin (50 mg/m ²)	Day 1	--	--	rest period	--	--	--	--	Rest period				
Prednisone (100 mg/m ²)	Day 1	Day 2	Day 3	Day 4	Day 5	--	--	--	Rest period				

*Dosing may continue for 2 more cycles (for a total of 8 cycles) if response is first seen at cycle 6.

Dose Modification Guidelines for Bortezomib When Given in Combination with Rituximab, Cyclophosphamide, Doxorubicin and Prednisone

Prior to the first day of each cycle (other than Cycle 1):

- Platelet count should be at least 100 x 10⁹/L and absolute neutrophil count (ANC) should be at least 1.5 x 10⁹/L.
 - Hemoglobin should be at least 8 g/dL (at least 4.96 mmol/L).
 - Non-hematologic toxicity should have recovered to Grade 1 or baseline.
- Interrupt bortezomib treatment at the onset of any Grade 3 hematologic or non-hematological toxicities, excluding neuropathy (see **Table 6** and **Special warnings and special precautions for use**). For dose adjustments, see **Table 5** below.

Table 5: Dose Modifications on Days 4, 8, and 11 during Cycles of Combination Bortezomib, Rituximab, Cyclophosphamide, Doxorubicin and Prednisone Therapy

Toxicity	Dose modification or delay
Hematological toxicity Grade 3 or higher neutropenia, or a platelet count not at or above 25 x 10 ⁹ /L	Withhold bortezomib therapy for up to 2 weeks until the patient has an ANC at or above 0.75 x 10 ⁹ /L and a platelet count at or above 25 x 10 ⁹ /L. • If, after bortezomib has been withheld, the toxicity does not resolve, discontinue bortezomib. • If toxicity resolves such that the patient has an ANC at or above 0.75 x 10 ⁹ /L and a platelet count at or above 25 x 10 ⁹ /L, bortezomib dose should be reduced by 1 dose level (from 1.3 mg/m ² to 1 mg/m ² , or from 1 mg/m ² to 0.7 mg/m ²).
Grade 3 or higher non-hematological toxicities	Withhold bortezomib therapy until symptoms of the toxicity have resolved to Grade 2 or better. Then, bortezomib may be reinitiated with one dose level reduction (from 1.3 mg/m ² to 1 mg/m ² , or from 1 mg/m ² to 0.7 mg/m ²). For bortezomib-related neuropathic pain and/or peripheral neuropathy, hold or modify bortezomib as outlined in Table 6.

For information concerning rituximab, cyclophosphamide, doxorubicin and prednisone, see manufacturer's prescribing information.

Dosage and Dose Modifications for Relapsed Multiple Myeloma and Relapsed Mantle Cell Lymphoma

Bortezomib (1.3 mg/m²/dose) is administered twice weekly for two weeks (Days 1, 4, 8, and 11) followed by a ten-day rest period (Days 12 to 21). For extended therapy of more than eight cycles, bortezomib may be administered on the standard schedule or, for relapsed multiple myeloma, on a maintenance schedule of once weekly for four weeks (Days 1, 8, 15, and 22) followed by a 13 day rest period (Days 23 to 35). At least 72 hours should elapse between consecutive doses of bortezomib.

Patients with multiple myeloma who have previously responded to treatment with bortezomib (either alone or in combination) and who have relapsed at least six months after their prior bortezomib therapy may be started on bortezomib at the last tolerated dose. Retreated patients are administered bortezomib twice weekly (Days 1, 4, 8, and 11) every three weeks for a maximum of eight cycles. At least 72 hours should elapse between consecutive doses of bortezomib. Bortezomib may be administered either as a single agent or in combination with dexamethasone.

Bortezomib therapy should be withheld at the onset of any Grade 3 non-hematologic or Grade 4 hematological toxicities excluding neuropathy as discussed below (see **Special warnings and special precautions for use**). Once the symptoms of the toxicity have resolved, bortezomib therapy may be reinitiated at a 25% reduced dose (1.3 mg/m²/dose reduced to 1 mg/m²/dose; 1 mg/m²/dose reduced to 0.7 mg/m²/dose).

For dose modifications guidelines for peripheral neuropathy, see Dose modifications for peripheral neuropathy below.

Dose Modifications for Peripheral Neuropathy

Starting bortezomib subcutaneously may be considered for patients with pre-existing or at high risk of peripheral neuropathy. When rituximab with pre-existing severe neuropathy should be treated with bortezomib only after careful risk-benefit assessment.

Patients experiencing new or worsening peripheral neuropathy during bortezomib therapy may require a decrease in the dose and/or a less dose-intense schedule.

For dose or schedule modification guidelines for patients who experience bortezomib-related neuropathic pain and/or peripheral neuropathy see **Table 6**.

Table 6: Recommended Dose Modification for Bortezomib related Neuropathic Pain and/or Peripheral Sensory or Motor Neuropathy

Severity of Peripheral Neuropathy Signs and Symptoms*	Modification of Dose and Regimen
Grade 1 (asymptomatic; loss of deep tendon reflexes or paresthesias) without pain or loss of function	No action
Grade 1 with pain or Grade 2 (moderate symptoms; limiting instrumental Activities of Daily Living (ADL)**)	Reduce bortezomib to 1 mg/m ²
Grade 2 with pain or Grade 3 (severe symptoms; limiting self care ADL***)	Withhold bortezomib therapy until toxicity resolves. When toxicity resolves reinitiate with a reduced dose of bortezomib at 0.7 mg/m ² once per week.
Grade 4 (life-threatening consequences; urgent intervention indicated)	Discontinue bortezomib

*Grading based on NCI Common Toxicity Criteria CTCv4.0.

**Instrumental ADL: refers to preparing meals, shopping for groceries or clothes, using telephone, managing money etc.

***Self-care ADL: refers to bathing, dressing and undressing, feeding self, using the toilet, taking medications, and not bedridden

Special Populations

Patients with Renal Impairment
The pharmacokinetics of bortezomib are not influenced by the degree of renal impairment. Therefore, dosing adjustments of bortezomib are not necessary for patients with renal insufficiency.

Since dialysis may reduce bortezomib concentrations, the drug should be administered after the dialysis procedure (see **Pharmacokinetic Properties**).

Patients with Hepatic Impairment

Patients with mild hepatic impairment do not require a starting dose adjustment and should be treated per the recommended bortezomib dose. For patients with moderate or severe hepatic impairment, see **Table 7** below, (also, see **Pharmacokinetic Properties**).

Table 7: Recommended Starting Dose Modification for Bortezomib in Patients with Hepatic Impairment

	Bilirubin Level	SGOT (AST) Levels	Modification of Starting Dose in Multiple Myeloma and Relapsed Mantle Cell Lymphoma (1.3 mg/m ² twice weekly)
Mild	≤1.0x ULN	≤1x ULN	None
Moderate	≤1.0x–1.5x ULN	Any	Reduce bortezomib to 0.7 mg/m ² in the first cycle.
Severe	>1.5x–3x ULN	Any	Consider dose escalation to 1.0 mg/m ² or further dose reduction to 0.5 mg/m ² in subsequent cycles based on patient tolerability.

Abbreviations: SGOT = serum glutamic oxaloacetic transaminase; AST = aspartate aminotransferase; ULN = upper limit of the normal range.

Administration

Bortezomib is administered intravenously or subcutaneously. When administered intravenously, bortezomib is administered as a 3-5 second bolus intravenous injection through a peripheral or central intravenous catheter followed by a flush with 0.9% sodium chloride solution for injection. For subcutaneous administration, the reconstituted solution is injected into the thighs (right or left) or abdomen (right or left). Injection sites should be rotated for successive injections.

If local injection site reactions occur following bortezomib injection subcutaneously, a less concentrated Bortezomib solution (1 mg/ml instead of 2.5 mg/ml) may be administered subcutaneously, or changed to IV injection.

4.3 Contraindications

Hypersensitivity to the active substance, to boron or to any of the excipients listed in section 6.1. Acute diffuse infiltrative pulmonary and pericardial disease.

When bortezomib is given in combination with other medicinal products, refer to their Summaries of Product Characteristics for additional contraindications.

4.4 Special warnings and precautions for use

When bortezomib is given in combination with other medicinal products, the Summary of Product Characteristics of these other medicinal products must be consulted prior to initiation of treatment with bortezomib. When thalidomide is used, particular attention to pregnancy testing and prevention requirements is needed (see section 4.6).

Intrathecal administration

There have been fatal cases of inadvertent intrathecal administration of bortezomib. Bortezomib Kabi 3.5 mg powder for solution for injection is for intravenous or subcutaneous use. Bortezomib should not be administered intrathecally.

Gastrointestinal toxicity

Gastrointestinal toxicity, including nausea, diarrhoea, vomiting and constipation are very common with bortezomib treatment. Cases of these have been uncommonly reported (see section 4.8). Therefore, patients who experience constipation should be closely monitored.

Haematological toxicity

Bortezomib treatment is very commonly associated with haematological toxicities (thrombocytopenia, neutropenia and anaemia). In studies in patients with relapsed multiple myeloma treated with bortezomib and in patients with previously untreated MCL treated with bortezomib in combination with rituximab, cyclophosphamide, doxorubicin, and prednisone (BR-CAP), one of the most common haematologic toxicity was transient thrombocytopenia.

Platelets were lowest at Day 11 of each cycle of bortezomib treatment and typically recovered to baseline by the next cycle. There was no evidence of cumulative thrombocytopenia. The mean platelet count nadir measured was approximately 40% of baseline in the single-agent multiple myeloma studies and 50% in the MCL study. In patients with advanced myeloma the severity of thrombocytopenia was related to pre-treatment platelet counts

